Mini Project 3 - Country Comparison

DATA CLEANING DOCUMENTATION

Scope: Power Query ETL & Data Preparation
Country Scope: ~263 Global Territories
Status: Completed & Validated

1. Executive Summary and Project Overview

This document details the end-to-end data cleaning, enrichment, and modeling workflow performed for the **Mini Project 3 - Country Comparison** project. The objective was to prepare a robust, error-free dataset covering ~263 countries to enable multidimensional comparisons across economic, demographic, energy, and infrastructure metrics.

2. RAW DATA INSPECTION & QUALITY ISSUES IDENTIFIED

DATA QUALITY ISSUES **Key Anomalies in Raw Source File**

- **Missing / Null Values across Numeric Metrics:** The raw source contained widespread missing values across critical numerical columns (e.g., oil consumption, gas reserves, birth rates, internet users).
- **Missing Region Dimension:** The primary dataset lacked a `Region` column necessary for aggregated regional analysis across the ~263 countries.
- **Data Type & Metadata Inconsistencies:** Multiple columns required standardizing data types (Text vs. Decimals/Whole Numbers) and cleaning text formatting across country names and header labels.

3. DATA CLEANING & TRANSFORMATION WORKFLOW (POWER QUERY)

STEP 1 **Source Investigation & Null Value Imputation**

Conducted extensive source checks to verify missing numerical values. Because secondary external sources were unavailable for these specific data gaps, all null values across numerical indicators were systematically replaced with 0 to preserve row counts, maintain mathematical integrity, and prevent DAX evaluation errors in Power BI.

STEP 2 **Custom Region Dimension Mapping & Merging**

Constructed an external mapping Excel file containing all ~263 countries and assigned their corresponding global `Region` classifications. Imported this lookup sheet into Power Query and performed a **Merge Queries** join on the `Country` column to enrich the core dataset with complete regional context.

STEP 3 **Data Type Enforcement & Text Standardization**

Applied explicit data type transformations across all columns (Decimal Number, Int64, Text). Cleaned textual metadata using Power Query transformations, including trimming leading/trailing whitespace, standardizing header names, and replacing obsolete string codes.

4. DATA MODELING & SCHEMA DESIGN

To optimize report performance, support multi-fact analysis, and facilitate clean DAX measure authoring, the dataset was structured into a **Galaxy Schema (Fact Constellation Schema)** comprising 1 Shared Dimension table and 4 Fact tables sharing `Dim Country` as a central conformed dimension:

Shared Conformed Dimension

- **Dim Country:** Primary central dimension containing `Country`, `Region`, `Development Level`, `GDP Category`, `Growth Indicator Category`, `Population Category`, and `Area(sq km)`.

1. Fact Demographic Health

- `Birth rate`, `Death rate`
- `Life expectancy at birth`
- `Infant mortality rate`
- `HIV/AIDS metrics`, `Population`

2. Fact Economic Metrics

- `GDP`, `GDP - per capita`
- `GDP - real growth rate(%)`
- `Exports`, `Imports`
- `Debt - external`, `Account balance`

3. Fact Energy Resources

- `Electricity consumption & production`
- `Natural gas consumption, exports & reserves`
- `Oil consumption & reserves`

4. Fact Infrastructure

- `Highways(km)`, `Railways(km)`, `Internet hosts`, `Internet users`, `Telephones - main lines in use`, `Telephones - mobile cellular`.

5. DATA VALIDATION & FINAL READINESS

VALIDATION **Quality Assurance Summary**

- **Galaxy Relationship Integrity:** Built 1-to-many bi-directional or single-direction relationships between the conformed `Dim Country[Country]` dimension and all 4 Fact tables.
- **Zero Mismatch:** Verified 100% match rate across all ~263 countries after the region merge.
- **DAX Stability:** Ensured zero null propagation issues across all 26 custom DAX measures.